

Hypoglykämie-Erkennung Explorationstudie

Robin van den Hoek

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Tabelle 1: Vergleich aller Modellvarianten zur Hypoglykämie-Erkennung (Cohortmean, 36 Patienten, Schwellenwert 70 mg/dL).

Gruppe	Modell	Precision	Recall	F1	F2
<i>Naive Baselines</i>					
	Naive Last Value	0.857	0.326	0.473	0.373
	Naive Mean	0.396	0.041	0.074	0.050
<i>Forecast-basierte Erkennung ($\tau = 60$ min)</i>					
	LSTM MSE	0.010	0.004	0.005	0.004
	LSTM Bounded-Lag	0.017	0.012	0.014	0.013
	LSTM Soft-DTW	0.039	0.009	0.014	0.010
	LSTM Multi-step	0.011	0.009	0.010	0.010
	PatchTST MSE	0.048	0.089	0.061	0.074
	PatchTST Bounded-Lag	0.055	0.095	0.066	0.080
	PatchTST Soft-DTW	0.056	0.129	0.062	0.084
	PatchTST Multi-step	0.060	0.103	0.072	0.085
<i>Direkte Event-Classifier (Any-time, 60 min)</i>					
	LSTM Event Classifier	0.237	0.693	0.334	0.463
	PatchTST Event Classifier	0.054	0.615	0.095	0.178
<i>Direkte Point-Event-Classifier (At exactly 60 min)</i>					
	LSTM Point Event Classifier	0.088	0.525	0.143	0.237
	PatchTST Point Event Classifier	0.036	0.537	0.064	0.124